

Sharks

PHYLUM	Chordata
CLASS	Chondrichthyes
SUBCLASS	Elasmobranchi (part)
ORDERS	9
FAMILIES	34
SPECIES	About 510

Sharks are formidable hunters with few natural predators when fully grown. Classic sharks have a streamlined body, powerful jaws with several rows of sharp teeth, and acute senses, especially smell. They range in size from the whale shark (the largest of all fishes) to squaloid sharks less than 12 in (30 cm) long. Sharks are most abundant in warm temperate and tropical waters, their habitats ranging from coastal areas to open oceans and deep basins. Some even venture into tropical rivers.

Anatomy

Most open-water sharks have a torpedo-shaped body that allows them to cut easily through water, although bottom-dwelling species are more varied and some have a flattened body, similar to that of rays. Unlike bony fishes, sharks have rigid fins that cannot be flexed or folded, yet their fins are nonetheless efficient for steering, propeling, and stabilizing the body during swimming. A shark's teeth, which are arranged

in rows, are replaced throughout its lifetime. The shape of the teeth varies according to diet (see p.473), although some species have 2 or more types of teeth. Sharks do not have a swim bladder, but their large, oil-filled liver adds to their buoyancy and helps prevent them from sinking. Many open-ocean sharks need to swim constantly to force water over their gills, but other species can pump water across their gills while motionless. Sharks

usually have 5 gill slits on either side of their head; a few species have 6 or 7 pairs.

Food and feeding

Sharks use various tactics to catch prey. Predatory sharks rely on short, high-speed pursuits through open water, or on ambush and surprise. Other species simply search an area for sedentary or slow-moving prey, or excavate burrowing animals. Some sharks are scavengers, using an acute sense of smell to locate carcasses (they can detect a scent in a solution as weak as one part per million). The 3 largest species of sharks are filter feeders, and at least one species, the cookie-cutter shark, is an ectoparasite on other sharks, large fishes, and marine mammals.

Reproduction

Coastal bays, lagoons, reefs, and atolls are the favored areas for female sharks to lay eggs or give birth (pupping). These areas provide a calm, food-rich environment for young sharks to grow and develop.

CONSERVATION

Over the last few decades, rising human populations and falling stocks of many food fishes have led to an increase in the popularity of sharks as a food source for humans. Compounding this pressure are the growth in the popularity of shark fishing as a sport and a rise in demand for shark fins in Asian cuisine. Millions of sharks are caught annually, of which blue shark and piked dogfish (*Squalus acanthias*) are two of the most heavily exploited. Conservation measures include restricting the number of fins that can be landed unattached, and protection of certain species and pupping grounds.

Baby sharks are born or hatch as miniature adults, even armed with fully functional teeth, and are ready to hunt. Sharks grow slowly to adulthood, often taking a decade or longer to reach maturity. During this time, many young sharks fall prey to larger predators.

HUNTING

Sharks of the open oceans (such as the blue shark, shown here) typically have a streamlined body and a powerful tail fin that propels them quickly through the water. The mainstays of their diet are squid and fishes. When hunting, they often circle their prey several times before attacking from below.

